

i) Recall the equation for the tangent plane at (x_0, y_0, z_0) .

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

$$\frac{\partial z}{\partial x} = \frac{1}{2\sqrt{4-x^2-2y^2}} \cdot -2x = -\frac{x}{z}$$

$$\text{at } p = (1, -1, 1): \frac{\partial z}{\partial x} = -1$$

$$\frac{\partial z}{\partial y} = \frac{1}{2\sqrt{4-x^2-2y^2}} \cdot -4y = -\frac{2y}{z}$$

$$\text{at } p = (1, -1, 1): \frac{\partial z}{\partial y} = 2$$

$$\Rightarrow \text{Tangent plane: } z - 1 = -(x - 1) + 2(y + 1)$$

$$z = -x + 2y + 4$$

$$\text{ii) } \frac{\partial z}{\partial x} = \frac{y}{x}, \quad \text{at } p = (1, 4, 0): \frac{\partial z}{\partial x} = 4$$

$$\frac{\partial z}{\partial y} = \log(x), \quad \text{at } p = (1, 4, 0): \frac{\partial z}{\partial y} = 0$$

$$\Rightarrow \text{Tangent plane: } z = 4(x - 1) = 4x - 4$$

$$\text{2 i) } dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

$$= 3x^2 \log(y^2) dx + \frac{2x^3 y}{y^2} dy$$

$$= 3x^2 \log(y^2) dx + \frac{2x^3}{y} dy$$

$$\text{ii) } \frac{\partial w}{\partial x} = \frac{1}{\sqrt{x^2+y^2+z^2}} \cdot \frac{1}{2\sqrt{x^2+y^2+z^2}} \cdot 2x$$

$$= \frac{x}{x^2+y^2+z^2}$$

$$\text{Similarly, } \frac{\partial w}{\partial y} = \frac{y}{x^2+y^2+z^2} \quad \text{and} \quad \frac{\partial w}{\partial z} = \frac{z}{x^2+y^2+z^2}$$

$$\Rightarrow dw = \frac{x}{x^2+y^2+z^2} dx + \frac{y}{x^2+y^2+z^2} dy + \frac{z}{x^2+y^2+z^2} dz$$

$$3 i) \quad \frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

$$\frac{\partial z}{\partial x} = \cos x \cos y = \cos(\pi t) \cos \sqrt{t}$$

$$\frac{\partial z}{\partial y} = -\sin x \sin y = -\sin(\pi t) \sin \sqrt{t}$$

$$\frac{dx}{dt} = \pi$$

$$\frac{dy}{dt} = \frac{1}{2\sqrt{t}}$$

$$\Rightarrow \frac{dz}{dt} = \pi \cos(\pi t) \cos \sqrt{t} - \frac{\sin(\pi t) \sin \sqrt{t}}{2\sqrt{t}}$$

$$ii) \quad \frac{\partial w}{\partial x} = y = e^t \sin t$$

$$\frac{\partial w}{\partial y} = x + z^2 = e^t + e^{2t} \cos^2 t$$

$$\frac{\partial w}{\partial z} = 2yz = 2e^{2t} \cos t \sin t$$

$$\frac{dx}{dt} = e^t$$

$$\frac{dy}{dt} = e^t \sin t + e^t \cos t$$

$$\frac{dz}{dt} = e^t \cos t - e^t \sin t$$

$$\begin{aligned} \Rightarrow \frac{dw}{dt} &= e^{2t} \sin t + (e^t + e^{2t} \cos^2 t) \{ e^t (\sin t + \cos t) \\ &\quad + 2e^{2t} \cos t \sin t (e^t (\cos t - \sin t)) \} \\ &= e^{2t} \sin t + e^{2t} \sin t + e^{2t} \cos t + e^{3t} \cos^2 t \sin t \\ &\quad + e^{3t} \cos^3 t + 2e^{3t} \cos^2 t \sin t + 2e^{3t} \cos t \sin^2 t \\ &= e^{2t} (2 \sin t + \cos t) + e^{3t} (3 \cos^2 t \sin t + \cos^3 t - 2 \cos t \sin^2 t) \end{aligned}$$

$$4 i) \quad \frac{\partial z}{\partial x} = 2x + y^3 = 31$$

$$\frac{\partial z}{\partial y} = 3xy^2 = 54$$

$$\frac{\partial x}{\partial u} = v^2 = 1$$

$$\frac{\partial y}{\partial u} = 1$$

$$\frac{\partial x}{\partial v} = 2uv = 4$$

$$\frac{\partial y}{\partial v} = e^w = 1$$

$$\frac{\partial x}{\partial w} = 3w^2 = 0$$

$$\frac{\partial y}{\partial w} = ve^w = 1$$

$$x = 2$$

$$y = 3$$

$$\frac{\partial z}{\partial u} = (2x + y^3) v^2 + 3xy^2 = 85$$

$$\frac{\partial z}{\partial v} = (2x + y^3) 2uv + 3xy^2 e^w = 178$$

$$\frac{\partial z}{\partial w} = (2x + y^3) \cdot 0 + 3xy^2 ve^w = 54$$

$$4.ii) \quad \frac{\partial R}{\partial u} = \frac{2u}{u^2+v^2+w^2} = \frac{6}{14} = \frac{3}{7}$$

$$\frac{\partial R}{\partial v} = \frac{2v}{u^2+v^2+w^2} = \frac{2}{14} = \frac{1}{7}$$

$$\frac{\partial R}{\partial w} = \frac{2w}{u^2+v^2+w^2} = \frac{4}{14} = \frac{2}{7}$$

$$\frac{\partial u}{\partial x} = 1 \quad \frac{\partial v}{\partial x} = 2 \quad \frac{\partial w}{\partial x} = 2y = 2$$

$$\frac{\partial u}{\partial y} = 2 \quad \frac{\partial v}{\partial y} = -1 \quad \frac{\partial w}{\partial y} = 2x = 2$$

$$u = x + 2y = 3 \quad v = 2x - y = 1 \quad w = 2xy = 2$$

$$\frac{\partial R}{\partial x} = \frac{\partial R}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial R}{\partial v} \frac{\partial v}{\partial x} + \frac{\partial R}{\partial w} \frac{\partial w}{\partial x} = \frac{3}{7} + \frac{2}{7} + \frac{4}{7} = \frac{9}{7}$$

$$\frac{\partial R}{\partial y} = \frac{\partial R}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial R}{\partial v} \frac{\partial v}{\partial y} + \frac{\partial R}{\partial w} \frac{\partial w}{\partial y} = \frac{6}{7} - \frac{1}{7} + \frac{4}{7} = \frac{9}{7}$$

5. in) Note that $x^2 + y^2 = r^2 \cos^2 \theta + r^2 \sin^2 \theta = r^2$
 and $\frac{y}{x} = \frac{r \sin \theta}{r \cos \theta} = \tan \theta \rightarrow \theta = \arctan \frac{y}{x}$

$$\Rightarrow \frac{\partial}{\partial x} r^2 = \frac{\partial}{\partial x} (x^2 + y^2)$$

$$2r \frac{\partial r}{\partial x} = 2x \Rightarrow \frac{\partial r}{\partial x} = \frac{x}{r} = \cos \theta$$

$$\text{Similarly, } \frac{\partial r}{\partial y} = \frac{y}{r} = \sin \theta$$

$$\frac{\partial \theta}{\partial x} = \frac{1}{1 + \left(\frac{y}{x}\right)^2} \cdot -\frac{y}{x^2} = \frac{-y}{x^2 + y^2} = -\frac{y}{r^2} = -\frac{\sin \theta}{r}$$

$$\frac{\partial \theta}{\partial y} = \frac{1}{1 + \left(\frac{y}{x}\right)^2} \cdot \frac{1}{x} = \frac{x}{x^2 + y^2} = \frac{x}{r^2} = \frac{\cos \theta}{r}$$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial z}{\partial \theta} \frac{\partial \theta}{\partial x} = \cos \theta \frac{\partial z}{\partial r} - \frac{\sin \theta}{r} \frac{\partial z}{\partial \theta}$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial z}{\partial \theta} \frac{\partial \theta}{\partial y} = \sin \theta \frac{\partial z}{\partial r} + \frac{\cos \theta}{r} \frac{\partial z}{\partial \theta}$$

$$\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = \cos^2 \theta \left(\frac{\partial z}{\partial r}\right)^2 - 2 \frac{\sin \theta \cos \theta}{r} \frac{\partial z}{\partial r} \frac{\partial z}{\partial \theta} + \frac{\sin^2 \theta}{r^2} \left(\frac{\partial z}{\partial \theta}\right)^2$$

$$+ \sin^2 \theta \left(\frac{\partial z}{\partial r}\right)^2 + 2 \frac{\sin \theta \cos \theta}{r} \frac{\partial z}{\partial r} \frac{\partial z}{\partial \theta} + \frac{\cos^2 \theta}{r^2} \left(\frac{\partial z}{\partial \theta}\right)^2$$

$$= \left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial z}{\partial \theta}\right)^2$$

$$5: b) \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\cos \theta \frac{\partial z}{\partial r} - \frac{y}{r^2} \frac{\partial z}{\partial \theta} \right)$$

$$= -\sin \theta \frac{\partial \theta}{\partial x} \frac{\partial z}{\partial r} + \cos \theta \left(\frac{\partial^2 z}{\partial r^2} \frac{\partial r}{\partial x} + \frac{\partial^2 z}{\partial r \partial \theta} \frac{\partial \theta}{\partial x} \right) + 2 \frac{y}{r^3} \frac{\partial r}{\partial x} \frac{\partial z}{\partial \theta} - \frac{y}{r^2} \left(\frac{\partial^2 z}{\partial \theta \partial r} \frac{\partial r}{\partial x} + \frac{\partial^2 z}{\partial \theta^2} \frac{\partial \theta}{\partial x} \right)$$

$$= \frac{\sin^2 \theta}{r} \frac{\partial z}{\partial r} + \cos^2 \theta \frac{\partial^2 z}{\partial r^2} - \frac{\sin \theta \cos \theta}{r} \frac{\partial^2 z}{\partial r \partial \theta} + \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial z}{\partial \theta} - \frac{\sin \theta \cos \theta}{r} \frac{\partial^2 z}{\partial \theta \partial r} + \frac{\sin^2 \theta}{r^2} \frac{\partial^2 z}{\partial \theta^2}$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(\sin \theta \frac{\partial z}{\partial r} + \frac{x}{r^2} \frac{\partial z}{\partial \theta} \right)$$

$$= \cos \theta \frac{\partial \theta}{\partial y} \frac{\partial z}{\partial r} + \sin \theta \left(\frac{\partial^2 z}{\partial r^2} \frac{\partial r}{\partial y} + \frac{\partial^2 z}{\partial r \partial \theta} \frac{\partial \theta}{\partial y} \right) - 2 \frac{x}{r^3} \frac{\partial r}{\partial y} \frac{\partial z}{\partial \theta} + \frac{x}{r^2} \left(\frac{\partial^2 z}{\partial \theta \partial r} \frac{\partial r}{\partial y} + \frac{\partial^2 z}{\partial \theta^2} \frac{\partial \theta}{\partial y} \right)$$

$$= \frac{\cos^2 \theta}{r} \frac{\partial z}{\partial r} + \sin^2 \theta \frac{\partial^2 z}{\partial r^2} + \frac{\sin \theta \cos \theta}{r} \frac{\partial^2 z}{\partial r \partial \theta} - \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial z}{\partial \theta} + \frac{\sin \theta \cos \theta}{r} \frac{\partial^2 z}{\partial \theta \partial r} + \frac{\cos^2 \theta}{r^2} \frac{\partial^2 z}{\partial \theta^2}$$

$$\Rightarrow \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \frac{1}{r} \frac{\partial z}{\partial r} + \frac{\partial^2 z}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2 z}{\partial \theta^2}$$

$$5: i) \quad s = \frac{x+y}{2} \quad t = \frac{x-y}{2}$$

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t} = \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}$$

$$\Rightarrow \frac{\partial z}{\partial s} \frac{\partial z}{\partial t} = \left(\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} \right) \left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} \right)$$

$$= \left(\frac{\partial z}{\partial x} \right)^2 - \left(\frac{\partial z}{\partial y} \right)^2$$

Question 6

$$i) \quad \|v\| = \sqrt{4^2 + (-3)^2} = 5 \quad \Rightarrow \quad \frac{v}{\|v\|} = \left(\frac{4}{5}, \frac{-3}{5}\right)$$

$$\nabla f = \left(2\sqrt{y}, \frac{x}{\sqrt{y}}\right)$$

$$D_v f(x, y) = \nabla f \cdot \frac{v}{\|v\|} = \frac{8}{5}\sqrt{y} - \frac{3}{5}\frac{x}{\sqrt{y}}$$

$$\text{At } P = (3, 4) : \frac{8}{5}\sqrt{4} - \frac{3}{5} \cdot \frac{3}{\sqrt{4}} = \frac{23}{10}$$

$$ii) \quad \|v\| = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14} \quad \Rightarrow \quad \frac{v}{\|v\|} = \left(\frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}\right)$$

$$\nabla f = \left(\frac{1}{y+z}, -\frac{x}{(y+z)^2}, -\frac{x}{(y+z)^2}\right)$$

$$D_v f(x, y) = \frac{1}{\sqrt{14}(y+z)} - \frac{2x}{\sqrt{14}(y+z)^2} - \frac{3x}{\sqrt{14}(y+z)^2} = \frac{y+z-5x}{\sqrt{14}(y+z)^2}$$

$$\text{At } P = (4, 1, 1) : \frac{1+1-5 \cdot 4}{\sqrt{14}(1+1)^2} = -\frac{9}{2\sqrt{14}}$$

Question 7

$$2) \quad v = (-2, -1, -3) \quad \|v\| = \sqrt{14} \quad \frac{v}{\|v\|} = \left(-\frac{2}{\sqrt{14}}, -\frac{1}{\sqrt{14}}, -\frac{3}{\sqrt{14}}\right)$$

$$\nabla f = (2x, 2y, 2z)$$

$$D_v f = -\frac{4x}{\sqrt{14}} - \frac{2y}{\sqrt{14}} - \frac{6z}{\sqrt{14}}$$

$$\text{At } P = (2, 1, 3) : \frac{-8-2-18}{\sqrt{14}} = \frac{-28}{\sqrt{14}} = -2\sqrt{14}$$

Question 8

$$3i) \quad \nabla f = (2xy^3z^4, 3x^2y^2z^4, 4x^2y^3z^3)$$

$$\text{At } P = (1, 1, 1) : \nabla f(P) = (2, 3, 4)$$

Maximal rate of change occurs in direction of

$$\nabla f(P) \text{ and equals } \|\nabla f(P)\| = \sqrt{2^2 + 3^2 + 4^2} = \sqrt{29}$$

$$\text{Direction : } \frac{\nabla f(P)}{\|\nabla f(P)\|} = \left(\frac{2}{\sqrt{29}}, \frac{3}{\sqrt{29}}, \frac{4}{\sqrt{29}}\right)$$

The minimal rate of change occurs in the opposite direction, i.e. $\left(-\frac{2}{\sqrt{29}}, -\frac{3}{\sqrt{29}}, -\frac{4}{\sqrt{29}}\right)$ and equals $-\sqrt{29}$.

3 ii) ~~$f(x, y, z) = \ln(x^1 y^2 z^3)$~~ ~~$f(x, y, z) = \ln(x^1 y^2 z^3)$~~ ~~$f(x, y, z) = \ln(x^1 y^2 z^3)$~~

$$f(x, y, z) = \ln(x y^2 z^3) = \ln(x) + 2\ln(y) + 3\ln(z)$$

$$\nabla f = \left(\frac{1}{x}, \frac{2}{y}, \frac{3}{z} \right)$$

$$\text{At } P = (1, -2, -3) : \nabla f(P) = (1, -1, -1)$$

$$\|\nabla f(P)\| = \sqrt{3}$$

Maximal rate of change: $\sqrt{3}$ in direction $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

Minimal rate of change: $-\sqrt{3}$ in direction $\left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

Question 9

4 i) $\nabla f = (2x, 4y, 6z)$

$$\nabla f(P) = \cancel{(2, 4, 6)} (8, -4, 6)$$

$\nabla f(P)$ is perpendicular to the tangent surface

→ The tangent plane is given by

$$\nabla f(P) \cdot (x - x_0, y - y_0, z - z_0) = 0$$

$$(8, -4, 6) \cdot (x - 4, y + 1, z - 1) = 0$$

$$8x - 3z - 4y - 4 + 6z - 6 = 0$$

$$8x - 4y + 6z = 10$$

ii) $f(x, y, z) = x - z - 4 \arctan(yz) = 0$

$$\nabla f(x, y, z) = \left(1, \frac{-4z}{1+(yz)^2}, -1 - \frac{4y}{1+(yz)^2} \right)$$

$$\nabla f(1+\pi, 1, 1) = (1, -2, -3)$$

The tangent plane is given by

$$(1, -2, -3) \cdot (x - (1+\pi), y - 1, z - 1) = 0$$

$$x - 1 - \pi - 2y + 2 - 3z + 3 = 0$$

$$x - 2y - 3z = \pi - 4$$

Question 10

5) Planes are parallel $\Leftrightarrow$ normal vectors are parallel.

$$\nabla(x^2 + 2y^2 + 3z^2) = (2x, 4y, 6z)$$

is a normal vector to the ellipsoid.

$\underline{n} = (3, -1, 3)$ is a normal vector to the plane $3x - y + 3z = 1$.

$\rightarrow$ We're looking for points (x, y, z) where
 $(2x, 4y, 6z) = c(3, -1, 3)$ for some c
 and $x^2 + 2y^2 + 3z^2 = 1$.

$$\rightarrow 2x = 3c \Rightarrow x = \frac{3c}{2}$$

$$4y = -c \Rightarrow y = -\frac{c}{4}$$

$$6z = 3c \Rightarrow z = \frac{c}{2}$$

Substituting into equation gives

$$\left(\frac{3c}{2}\right)^2 + 2\left(-\frac{c}{4}\right)^2 + 3\left(\frac{c}{2}\right)^2 = 1$$

$$\frac{9}{4}c^2 + \frac{1}{8}c^2 + \frac{3}{4}c^2 = 1$$

$$\frac{25}{8}c^2 = 1 \Rightarrow c = \pm \sqrt{\frac{8}{25}} = \pm \frac{2}{5}\sqrt{2}$$

$c = \frac{2}{5}\sqrt{2}$ corresponds to the point $\left(\frac{3}{5}\sqrt{2}, -\frac{1}{10}\sqrt{2}, \frac{1}{5}\sqrt{2}\right)$

$c = -\frac{2}{5}\sqrt{2}$ corresponds to $\left(-\frac{3}{5}\sqrt{2}, \frac{1}{10}\sqrt{2}, -\frac{1}{5}\sqrt{2}\right)$